

Yunfei Ge

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EDUCATION

Northwestern University, Chicago, IL, USA

Sep. 2025 – Jun. 2027

M.S. in Computer Engineering | GPA 4.0/4.0

Shanghai University, Shanghai, China

Sep. 2021 – Jun. 2025

B.Eng. in Computer Science and Technology | Freshman Scholarship (Top 5%)

PUBLICATIONS

1. [2026 | NeurIPS 2026 (Under Review)] **Yunfei Ge***, Qineng Wang*, et al., Jiajun Wu, Manling Li *MindTopo: Can Foundation Models Reason in Topological Space?*
2. [2026 | Preprint] Xinyu Che*, Junqi Xiong*, **Yunfei Ge***, et al. *MMG2Skill: Can Agents Distill In-the-Wild Guides into Self-Evolving Skills?*

RESEARCH EXPERIENCE

MLL Lab, Northwestern University

Sep. 2025 – Present

Research Assistant **Topic: 3D Spatial Reasoning | VLM Benchmark | Adv. Prof. Manling Li**

1. As first author, designed the MindTopo benchmark to measure topological intuition as **one complete cognitive chain**: a unified task schema formalizes 5 Piagetian primitives at both the reasoning and planning levels — single-turn visual-QA tuples and Gym-style MDP rollouts — so each primitive is first recognized, then operated on, as consecutive links of one chain rather than two separate test sets; this chain is what exposes the action gap: models recognize a topological invariant statically yet cannot maintain it across an action sequence (best model GPT-5.5 54.13% vs human 97.40%).
2. Built the measurement rig behind it: a 4-stage procedural-generation pipeline producing 13 task types and 11,016 difficulty-controlled instances with ground-truth labels by construction, a deterministic reproducible batched harness over 11 frontier VLMs, and a 7-class pipeline-aligned error taxonomy that pinpoints where the chain breaks (reasoning errors concentrate at early perception, planning errors at late action); underneath, a self-built Blender Knot plugin drives a Gauss-code Blender → MuJoCo physics-simulation and rendering pipeline as the topology scene/simulation infrastructure.

Nanjing University LINKLAB

Apr. 2026 – Jun. 2026

Research Assistant **Topic: VLM Agents | Guide-to-Skill Learning | Adv. Prof. Jiaheng Liu**

1. Core contributor to MMG2Skill — a closed-loop framework that distills in-the-wild multimodal Web guides into editable, self-evolving skills: a VLM extractor converts guides into structured skills, conditions a fixed VLM agent during rollout, and iteratively refines skills through analyzer-driven trajectory diagnosis and reviser updates.
2. Integrated the external EmbodiedEval benchmark into the MMG2Skill evaluation framework: adapted its interactive 3D-scene tasks (navigation, object interaction, spatial QA) to a unified task schema and rollout protocol, and wired the VLM agent's observation/action interface, success metrics, and error-analysis flow.

Scale Lab, Shanghai Jiao Tong University

Apr. 2025 – Jul. 2025

Research Assistant **Topic: Bimanual Manipulation | VLA Benchmark | Adv. Prof. Yao Mu**

1. Contributed to task extension and simulation debugging for the RoboTwin 2.0 bimanual manipulation benchmark: configured collaborative grasping, transfer, and placement tasks with SAPIEN / RoboTwin task APIs; added task configs for cross-embodiment data collection and evaluation.
2. Integrated the RoboTwin 2.0 data-generation workflow and applied domain randomization to diversify scenes; wrote trajectory cleaning and format-conversion scripts to organize HDF5 observation-action data into expert trajectories for imitation learning and policy evaluation.

PROFESSIONAL EXPERIENCE

Wegwiser (Boston, USA)

Nov. 2025 – Apr. 2026

AI Engineer Intern **Topic: RAG & LLM Application**

1. Rebuilt the product QA system with a LangGraph Supervisor + multi-specialist agent architecture, where a central router handles intent classification and dynamic routing while specialist agents use dedicated tool chains and ReAct-style reasoning.
2. Built a dual-pathway RAG workflow for strategic-alignment and factual-retrieval questions; also developed an AI design-review module and an EWMA-weighted skill-to-task matching model for product consulting workflows.

Duo LuoLuo (Shanghai)

Apr. 2025 – Aug. 2025

AI Engineer Intern **Topic: LLM Fine-tuning & Edge Device Optimization**

1. Built a Qwen2.5 fine-tuning pipeline with data validation over 4K synthetic samples; integrated Elasticsearch-based real-time RAG retrieval.
2. Optimized the inference stack with TensorRT, ONNX Runtime, and INT8 PTQ quantization.

SAIC Maxus (Shanghai)

Nov. 2023 – May 2024

Software Engineer Intern **Topic: Microservices & System Architecture**

1. Contributed to refactoring a monolith into Spring Boot and Docker-based microservices; automated industrial data exchange with Kafka async scheduling and RESTful APIs.

COMPETITION

RoboMaster Robotics Competition National Third Prize

Sep. 2021 – Apr. 2023

Vision Group Member **Topic: Computer Vision & Autonomous Navigation**

1. Contributed to the auto-aim vision module: trained a YOLOv5 armor-plate keypoint detector, used PnP pose estimation to compute gimbal offsets, and sent commands to the embedded controller via serial.
2. Led autonomous navigation for the infantry robot using 2D LiDAR, Cartographer SLAM, and ROS Navigation for real-time mapping, global path planning, and local obstacle avoidance.

TECHNICAL SKILLS

Languages: Python, C++, Java

LLM/Agent: PyTorch, Transformers, LLaMA-Factory, LoRA / QLoRA, GRPO / PPO, RAG

Inference & Serving: TensorRT, ONNX Runtime, vLLM, INT8 / PTQ quantization

Robotics/Sim: ROS, MuJoCo, Isaac Sim, Blender, SolidWorks

Infrastructure: Git, Linux, Docker, K8s, Kafka, Spring Boot